

DEEP OPERATOR LEARNING OF MAXIMUM-PRINCIPLE-PRESERVING DISSIPATIVE SEMIGROUPS FOR TIME-DEPENDENT PDES

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Abstract. We formulate operator learning for autonomous evolution PDEs at the semigroup level: rather than learning only a one-step predictor, the learner is a family $\Phi_\theta(\cdot, \tau)$ intended for repeated composition. We derive discrete stability estimates showing how trajectory-level approximation error and composition defect accumulate. We then construct a spatially finite-dimensional, temporally continuous latent dissipative architecture for which interior-state invariance, exact composition of the continuous flow, and dissipation of an induced learned energy hold conditionally on global existence. A revised transfer theorem separates the exact semidiscrete flow, the exact latent flow, spatial reconstruction, and numerical time integration; a componentwise error bound replaces the previous unqualified universality claim. Archived benchmark values are retained as pre-revision results, but they require rerunning under strict time alignment before supporting comparative accuracy claims. Fisher–KPP and Allen–Cahn are in-class reaction–diffusion examples, whereas viscous Burgers is reported only as an out-of-class transport stress test.

1. Introduction and semigroup formulation. We study time-dependent PDEs whose solution operators form nonlinear semigroups and satisfy structural laws such as invariant-region preservation and Lyapunov dissipation. These laws are central both to the mathematical interpretation of the dynamics and to the stability of long-time repeated composition. Let X be a Banach space of states, for example $L^\infty(\Omega)$, $L^2(\Omega)$, or a Sobolev space on a spatial domain $\Omega \subset \mathbb{R}^d$. Consider an autonomous evolution PDE of the form

$$(1.1) \quad \partial_t u = \mathcal{F}(u), \quad t > 0, \quad u(0) = u_0 \in X.$$

Assume that (1.1) is well-posed [32] and generates a nonlinear solution semigroup

$$(1.2) \quad \mathcal{S}_t : X \rightarrow X, \quad u(t) = \mathcal{S}_t u_0, \quad t \geq 0.$$

Thus

$$(1.3) \quad \mathcal{S}_0 = \text{Id}_X, \quad \mathcal{S}_{t+s} = \mathcal{S}_t \circ \mathcal{S}_s, \quad t, s \geq 0.$$

Our focus is on problems for which the semigroup satisfies two additional structural properties.

Maximum principle / invariant region.. There exists a closed admissible set $\mathcal{K} \subset X$ such that

$$(1.4) \quad \mathcal{S}_t(\mathcal{K}) \subset \mathcal{K}, \quad t \geq 0.$$

A typical example is

$$(1.5) \quad \mathcal{K} = \{u \in X : m \leq u(x) \leq M \text{ a.e. in } \Omega\},$$

which encodes a maximum principle, positivity, or boundedness property.

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Dissipativity. There exists a Lyapunov functional

$$(1.6) \quad \mathcal{E} : X \rightarrow \mathbb{R} \cup \{+\infty\}$$

for which

$$(1.7) \quad \mathcal{E}(S_t u) \leq \mathcal{E}(u), \quad t \geq 0, \quad u \in \mathcal{K}.$$

Depending on the PDE, $\mathcal{E}$ may be an energy, entropy, free energy, or another dissipated quantity.

Standard operator-learning formulations often emphasize short-time prediction. Our focus is different: we seek a learned evolution family intended for repeated composition in time. The approximation target is therefore the operator family $(S_t)_{t \geq 0}$ itself. This emphasis matters because a learned one-step map may fit short-time data and yet accumulate structural errors under repeated composition: small violations of invariance, dissipation, or semigroup consistency can be amplified when the model is used as a time-marching solver. The problem we address is to construct a learned family $(\Phi_\theta(\cdot, \tau))_{\tau \geq 0}$ that approximates $(S_t)_{t \geq 0}$ while retaining (1.4) and (1.7). In other words, the objective is not only local prediction accuracy but a learned evolution family that can be composed in time without destroying the PDE’s structural laws.

Contributions. This paper makes four contributions. First, it formulates structure-preserving operator learning at the nonlinear semigroup level: rather than learning an isolated one-step predictor, the target is an evolution family $\Phi_\theta(\cdot, \tau)$ intended for repeated composition, with the PDE semigroup $(S_t)_{t \geq 0}$ as the reference object (Definition 3.1). Second, it derives error-propagation estimates that decompose the two sources of error under repeated composition — trajectory-level approximation and semigroup defect — showing that both enter discrete time-stepping stability in complementary ways (Section 4). Third, it constructs a semidiscrete latent dissipative architecture (Sections 5–6) for which, assuming the latent ODE generates a global flow, invariance of the interior state set $(m, M)^N$, exact semigroup composition of the continuous flow, and dissipation of an induced learned energy hold by construction. Fourth, Theorem 8.1 gives a conditional exact-flow comparison with explicit projection and reconstruction maps, while Proposition 8.2 adds numerical integration errors separately. Proposition 9.1 supplies a proved error bound for architecture-compatible generators; density of the implemented restricted component classes remains open. The layered character of the framework is deliberate: PDE, exact semidiscrete, exact latent, numerical, and empirical statements are not interchangeable.

Section 2 positions the present work relative to the literature. Section 4 develops the discrete stability and error-propagation analysis. Sections 5–6 construct the latent architecture and its neural parametrization. Section 7 documents numerical experiments. Sections 8–9 give the conditional transfer and architecture-compatible component bounds.

2. Relation to existing work. The present formulation is connected to several lines of work without coinciding with any single one of them. We organize the discussion around five themes.

Nonlinear semigroup theory. The mathematical foundation for autonomous evolution PDEs is the theory of nonlinear semigroups [9, 11, 10]. The standard setup associates a closed operator $\mathcal{F}$ on a Banach space X with a solution semigroup $(S_t)_{t \geq 0}$ satisfying $S_0 = \text{Id}$ and $S_{t+s} = S_t \circ S_s$. Structural laws such as maximum principles

and Lyapunov dissipation are classical features of this framework. Our paper takes these structural properties as the target specification for the learned model, rather than treating them as optional training penalties.

Structure-preserving numerical methods.. The idea that numerical discretizations should inherit structural properties of the continuous PDE has a long history in numerical analysis. Invariant-region preserving schemes guarantee that the discrete solution remains in a physically admissible set [16, 15, 29]. Strong stability preserving (SSP) time integrators [26] preserve convex constraints and total variation bounds. Geometric numerical integration [25, 24] targets symplectic, volume-preserving, or energy-conserving properties. Implicit-explicit (IMEX) methods [17] separate stiff from non-stiff terms to maintain stability. Our approach differs in that the structural guarantees are built into the architecture of the learned evolution family, rather than into a fixed numerical scheme applied to a known vector field.

Deep learning for PDEs.. The use of neural networks for PDE problems was catalyzed by the dynamical systems perspective of E [13] and by physics-informed neural networks (PINNs) [12]. Early work on neural-network-based ODE and PDE solvers [33] demonstrated the feasibility of the approach, and high-dimensional PDE solvers based on deep learning [34] extended the scope to previously intractable problems. Deep Galerkin methods [14] and neural-network-based PDE solvers [23] have since expanded the scope of neural approaches. Our work differs from these in target: rather than solving a single PDE instance, we seek to learn the solution operator family itself, as in operator learning approaches.

Operator learning and neural operators.. DeepONet [6, 30] and the Fourier neural operator (FNO) [5] approximate input-output maps between function spaces. The general neural operator framework [4] provides universality results for such approximations. These approaches typically learn a single-step map; our emphasis is on a whole evolution family intended for repeated composition in time, with structural guarantees that persist under iteration.

Structure-preserving neural ODEs.. Neural ODEs [1, 20, 35] parametrize continuous-time dynamics with neural networks. Hamiltonian neural networks (HNNs) [2] and SympNets [3] encode symplectic structure. OnsagerNet [7] enforces dissipative structure via the Onsager principle. Port-Hamiltonian neural networks [8] encode dissipative structure through interconnection and damping, and Lyapunov-based neural approaches construct stability certificates for learned dynamics. Symplectic recurrent neural networks [18] and stable neural flows [19] address long-time stability of learned dynamics. Invariant-region-preserving neural networks [21] are the closest in spirit to the present paper. Recent work on the interplay between discretization and regularization in neural ODEs [22] further highlights the importance of understanding how learned dynamics behave under repeated composition. Our contribution adds the semigroup-level formulation and the explicit error-propagation analysis under repeated composition, while the architectural construction (sigmoid logit map + gradient flow with PSD mobility) shares the general principle of encoding constraints via diffeomorphisms to unconstrained spaces.

The distinction of the present paper is that it uses the semigroup viewpoint to keep time composition, invariant-region preservation, and energy decay in the same target specification, while leaving continuous-to-discrete PDE transfer as a separate analytical issue.

3. Target class of learned semigroups. We next specify the class of learned families that is admissible for this objective. The basic mathematical object is a

parameterized family of learned operators

$$(3.1) \quad \Phi_\theta(\cdot, \tau) : \mathcal{K} \rightarrow \mathcal{K}, \quad \tau \geq 0,$$

intended to approximate the solution operator $\mathcal{S}_\tau$. The five requirements below fall into two groups: (3.2)–(3.4) concern approximation of the evolution family itself, whereas (3.5)–(3.6) guarantee that the learned repeated composition stays inside the admissible set and continues to dissipate the chosen Lyapunov functional.

DEFINITION 3.1 (Structure-preserving deep semigroup learner). *A family $\Phi_\theta(\cdot, \tau) : \mathcal{K} \rightarrow \mathcal{K}$ is called a structure-preserving deep semigroup learner if it satisfies the following properties.*

(S1) **Identity at zero time:**

$$(3.2) \quad \Phi_\theta(u, 0) = u, \quad u \in \mathcal{K}.$$

(S2) **Generator consistency:** for $u \in D(\mathcal{F}) \cap \mathcal{K}$,

$$(3.3) \quad \Phi_\theta(u, \tau) = u + \tau \mathcal{F}(u) + o(\tau), \quad \tau \downarrow 0.$$

(S3) **Semigroup defect control:** there exists a defect function $\eta_{\text{sg}} : [0, \infty)^2 \rightarrow [0, \infty)$ such that

$$(3.4) \quad \|\Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) - \Phi_\theta(u, \tau_1 + \tau_2)\| \leq \eta_{\text{sg}}(\tau_1, \tau_2),$$

for all $u \in \mathcal{K}$ and all $\tau_1, \tau_2 \geq 0$, with $\eta_{\text{sg}}(\tau_1, \tau_2) \rightarrow 0$ as $(\tau_1, \tau_2) \rightarrow (0, 0)$.

(S4) **Maximum-principle preservation:**

$$(3.5) \quad u \in \mathcal{K} \implies \Phi_\theta(u, \tau) \in \mathcal{K}, \quad \tau \geq 0.$$

(S5) **Dissipation preservation:**

$$(3.6) \quad \mathcal{E}(\Phi_\theta(u, \tau)) \leq \mathcal{E}(u), \quad u \in \mathcal{K}, \quad \tau \geq 0.$$

Definition 3.1 isolates the class of learned operator families that are admissible for the present problem. The first three clauses encode approximation of the PDE semigroup, whereas (3.5) and (3.6) encode preservation of the maximum principle and of the dissipation law. In particular, (3.4) measures whether the learned family behaves coherently across different time increments, while (3.5)–(3.6) are the structural conditions that prevent rollout from leaving the physically meaningful regime. This definition therefore serves as the interface between the continuous semigroup problem and the architectural constructions introduced later. It should be read as a PDE-level target specification; the latent architecture constructed later verifies only a finite-dimensional structural subset of these requirements, while generator matching and approximation to the target PDE semigroup remain separate issues.

REMARK 3.1 (FINITE-DIMENSIONAL ANALOGUE). For the semidiscrete model considered later, the PDE-level Definition 3.1 has a finite-dimensional counterpart. Let $X_N = \mathbb{R}^N$, $\mathcal{K}_N = [m, M]^N$, and let $\mathcal{F}_N : \mathcal{K}_N \rightarrow \mathbb{R}^N$ be a semidiscrete generator. A family $\Phi_\theta(\cdot, \tau) : \mathcal{K}_N \rightarrow \mathcal{K}_N$ is a *finite-dimensional structure-preserving learner* if it satisfies (3.2) on $\mathcal{K}_N$, the finite-dimensional generator relation

$$(3.7) \quad \Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad u \in \mathcal{K}_N, \quad \tau \downarrow 0,$$

and (3.4)–(3.6) with $\mathcal{K}$ replaced by $\mathcal{K}_N$ and $\mathcal{E}$ replaced by a discrete energy $\mathcal{E}_N : \mathcal{K}_N \rightarrow \mathbb{R}$. The latent architecture constructed in Sections 5–6 verifies this finite-dimensional version on the interior set $(m, M)^N$, as detailed in Corollary 5.7. The passage from the finite-dimensional learner back to the PDE-level specification is the subject of Problem 10.4 in Section 10.

4. Repeated composition and discrete stability. Fix a time step $\Delta t > 0$.

Once the family Φ_θ is learned, it generates a discrete solver by repeated composition:

$$(4.1) \quad u^{n+1} = \Phi_\theta(u^n, \Delta t), \quad u^0 = u_0,$$

or equivalently

$$(4.2) \quad u^n = (\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0), \quad n \in \mathbb{N} \cup \{0\}.$$

This section clarifies what is gained by learning the semigroup rather than only a one-step map. Once the model is used for repeated time-stepping, two questions become central: does the discrete orbit remain in the admissible set and dissipate the Lyapunov functional, and how do one-step approximation error and semigroup inconsistency accumulate over many compositions?

THEOREM 4.1 (Discrete stability inherited from structure). *Assume that (3.5) and (3.6) hold. Then, for every $u_0 \in \mathcal{K}$, the discrete orbit (4.2) satisfies*

$$(4.3) \quad u^n \in \mathcal{K}, \quad \mathcal{E}(u^{n+1}) \leq \mathcal{E}(u^n) \leq \mathcal{E}(u^0), \quad n \in \mathbb{N} \cup \{0\}.$$

In particular, the learned solver preserves the admissible set and satisfies a discrete Lyapunov law.

Proof. The proof is an induction on n . Since $u^0 = u_0 \in \mathcal{K}$, the base step is immediate. If $u^n \in \mathcal{K}$, then (3.5) gives $u^{n+1} = \Phi_\theta(u^n, \Delta t) \in \mathcal{K}$, and (3.6) yields

$$\mathcal{E}(u^{n+1}) = \mathcal{E}(\Phi_\theta(u^n, \Delta t)) \leq \mathcal{E}(u^n).$$

This proves (4.3). $\square$

Theorem 4.1 shows that the structural constraints are not merely formal conditions: they immediately turn the learned operator into a stable time-marching scheme. In particular, once (3.5) and (3.6) are built into the model, admissibility and Lyapunov decay propagate automatically to every iterate.

PROPOSITION 4.2 (Error decomposition through semigroup defect). *Let $T > 0$ and $u_0 \in \mathcal{K}$, and assume that (3.4) and (3.5) hold. Assume also that there exists $L_T \geq 0$ such that*

$$(4.4) \quad \|\Phi_\theta(v, \tau) - \Phi_\theta(w, \tau)\| \leq (1 + L_T \tau) \|v - w\|, \quad 0 \leq \tau \leq T,$$

for all $v, w \in \mathcal{K}$. Define

$$(4.5) \quad \varepsilon_{\text{app}}(T; u_0) := \sup_{0 \leq t \leq T} \|\Phi_\theta(u_0, t) - \mathcal{S}_t u_0\|,$$

and

$$(4.6) \quad \varepsilon_{\text{sg}}(T, \Delta t) := \sup_{0 \leq t \leq T - \Delta t} \eta_{\text{sg}}(t, \Delta t).$$

Then, for every integer n with $0 \leq n\Delta t \leq T$,

$$(4.7) \quad \left\| (\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0) - \mathcal{S}_{n\Delta t} u_0 \right\| \leq e^{L_T T} n \varepsilon_{\text{sg}}(T, \Delta t) + \varepsilon_{\text{app}}(T; u_0).$$

Proof. Set

$$(4.8) \quad \delta_n := \left\| (\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0) - \Phi_\theta(u_0, n\Delta t) \right\|.$$

Then $\delta_0 = 0$. By (3.5), the iterated state $(\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0)$ remains in $\mathcal{K}$; moreover $\Phi_\theta(u_0, n\Delta t) \in \mathcal{K}$. Hence the Lipschitz estimate applies to the two arguments below. For $n\Delta t \leq T - \Delta t$,

$$\begin{aligned} \delta_{n+1} &\leq \left\| \Phi_\theta((\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0), \Delta t) - \Phi_\theta(\Phi_\theta(u_0, n\Delta t), \Delta t) \right\| \\ &\quad + \left\| \Phi_\theta(\Phi_\theta(u_0, n\Delta t), \Delta t) - \Phi_\theta(u_0, (n+1)\Delta t) \right\| \\ &\leq (1 + L_T \Delta t) \delta_n + \varepsilon_{\text{sg}}(T, \Delta t). \end{aligned}$$

A discrete Gronwall argument gives the displayed bound below; we use the coarse estimate $\sum_{j=0}^{n-1} (1 + L_T \Delta t)^j \leq n e^{L_T T}$, which is sufficient for the convergence statement.

$$(4.9) \quad \delta_n \leq e^{L_T T} n \varepsilon_{\text{sg}}(T, \Delta t).$$

Combining (4.9) with the triangle inequality yields (4.7). $\square$

Proposition 4.2 separates two conceptually different error sources. The term ε_{app} measures the trajectory-level semigroup fit from the initial state u_0 , whereas ε_{sg} measures the internal inconsistency of the learned composition law. The estimate shows that long-time repeated composition is controlled not only by short-time approximation quality, but also by how well the learned family behaves as a semigroup. This bound is conditional on the stability hypothesis (4.4) and does not by itself provide a PDE-level approximation theorem.

REMARK 4.1 (DOMAIN OF THE LIPSCHITZ CONDITION). Proposition 4.2 assumes (4.4) holds on the full admissible set $\mathcal{K}$. In practice, the Lipschitz verification (Proposition 6.4 below) establishes the condition only on compact subsets $\mathcal{U} \subset (m, M)^N$, because g^{-1} is Lipschitz only on compact subsets of the interior. The error bound (4.7) therefore applies to initial data u_0 whose trajectory under Φ_θ remains in a compact subset of the interior for $0 \leq t \leq T$. This restriction is natural: states near the boundary $\{u_i = m \text{ or } u_i = M\}$ have $g^{-1}(u_i) \rightarrow \pm\infty$, and the latent dynamics may degenerate there.

COROLLARY 4.3 (A route to convergence). *Let $T > 0$ and $u_0 \in \mathcal{K}$. Assume that, for a sequence $\Delta t_m \downarrow 0$ and corresponding learned models satisfying (3.4)–(3.5), one has*

$$(4.10) \quad \varepsilon_{\text{app}}^{(m)}(T; u_0) \rightarrow 0, \quad \frac{\varepsilon_{\text{sg}}^{(m)}(T, \Delta t_m)}{\Delta t_m} \rightarrow 0,$$

and that (4.4) holds with a constant L_T independent of m . Then

$$(4.11) \quad \sup_{0 \leq n\Delta t_m \leq T} \left\| (\Phi_\theta^{(m)}(\cdot, \Delta t_m))^{\circ n}(u_0) - \mathcal{S}_{n\Delta t_m} u_0 \right\| \rightarrow 0$$

as $m \rightarrow \infty$.

Proof. If $0 \leq n\Delta t_m \leq T$, then $n \leq T/\Delta t_m$. Proposition 4.2 gives

$$\left\| (\Phi_\theta^{(m)}(\cdot, \Delta t_m))^{\circ n}(u_0) - \mathcal{S}_{n\Delta t_m} u_0 \right\| \leq e^{L_T T} T \frac{\varepsilon_{\text{sg}}^{(m)}(T, \Delta t_m)}{\Delta t_m} + \varepsilon_{\text{app}}^{(m)}(T; u_0).$$

Taking the supremum over n and using (4.10) proves (4.11). $\square$

5. A latent structure-preserving architecture. Corollary 4.3 therefore suggests a concrete design principle: a learned evolution family should control both approximation error and semigroup defect, while preserving the structural laws that guarantee stable repeated composition. The next question is how to build these requirements into an architecture for which the structural clauses of Definition 3.1 hold at the model-class level rather than only through soft penalties during training.

We now work with a semidiscrete finite-dimensional state model (Assumption 5.1 below). The exact statements proved in the remainder of the paper are therefore architecture-level finite-dimensional guarantees, not direct PDE-level transfer results. The rigorous passage from this exact finite-dimensional verification back to the target infinite-dimensional semigroup is deferred to Section 10.

ASSUMPTION 5.1 (FINITE-DIMENSIONAL STATE MODEL). *Fix $N \geq 1$, and identify the spatially discretized state space with*

$$(5.1) \quad X_N = \mathbb{R}^N.$$

Assume that the maximum principle is represented by the box constraint

$$(5.2) \quad \mathcal{K}_N = [m, M]^N = \{u = (u_1, \dots, u_N)^\top \in \mathbb{R}^N : m \leq u_i \leq M, i = 1, \dots, N\}.$$

We regard X_N and $\mathcal{K}_N$ as the finite-dimensional counterparts of X and $\mathcal{K}$, respectively. For the inverse-logit latent construction used below, the actual input domain is the interior set $(m, M)^N$, because the latent inverse g^{-1} is only defined there. The resulting learned family is therefore considered as a map

$$(5.3) \quad \Phi_\theta : (m, M)^N \times [0, \infty) \rightarrow \mathcal{K}_N.$$

The construction below follows a simple architectural principle: first lift an interior constrained state into an unconstrained latent variable, then evolve that latent variable by an autonomous dissipative flow, and finally map the latent state back into the admissible set. Boundary states with some component equal to m or M are not covered by the present parametrization.

DEFINITION 5.1 (Bounded latent parametrization). *Let $\sigma : \mathbb{R} \rightarrow (0, 1)$ be the sigmoid function*

$$(5.4) \quad \sigma(\xi) = \frac{1}{1 + e^{-\xi}}.$$

Define $g : \mathbb{R}^N \rightarrow (m, M)^N$ componentwise by

$$(5.5) \quad g(z)_i = m + (M - m)\sigma(z_i), \quad i = 1, \dots, N.$$

Then g maps the whole latent space into the admissible set. Its inverse on $(m, M)^N$ is

$$(5.6) \quad g^{-1}(u)_i = \log\left(\frac{u_i - m}{M - u_i}\right), \quad i = 1, \dots, N,$$

and its Jacobian is

$$(5.7) \quad Dg(z) = (M - m) \operatorname{diag}(\sigma(z_i)(1 - \sigma(z_i)))_{i=1}^N.$$

DEFINITION 5.2 (Latent dissipative semigroup). Assume that

$$(5.8) \quad \Psi_\theta \in C^1(\mathbb{R}^N; \mathbb{R}), \quad K_\theta : \mathbb{R}^N \rightarrow \mathbb{R}^{N \times N},$$

and that $K_\theta(z)$ is symmetric positive semidefinite for every $z \in \mathbb{R}^N$. Consider the latent ODE

$$(5.9) \quad \partial_s Z_\theta(s; z_0) = -K_\theta(Z_\theta(s; z_0)) \nabla \Psi_\theta(Z_\theta(s; z_0)), \quad Z_\theta(0; z_0) = z_0.$$

Assume furthermore that this vector field generates a local flow φ_θ^τ . The learned operator family is then defined for interior states and for those times $\tau \geq 0$ for which the latent trajectory through $g^{-1}(u)$ exists, by

$$(5.10) \quad \Phi_\theta(u, \tau) := g(\varphi_\theta^\tau(g^{-1}(u))).$$

The induced discrete energy on $(m, M)^N$ is

$$(5.11) \quad \mathcal{E}_\theta(u) := \Psi_\theta(g^{-1}(u)).$$

REMARK 5.1 (LEARNED, DISCRETE, AND PHYSICAL ENERGIES). The induced functional $\mathcal{E}_\theta$ is a learned Lyapunov function. It is not automatically equal to a discrete physical energy $\mathcal{E}_N$, nor does its monotonicity imply monotonicity of the PDE functional $\mathcal{E}$. Establishing $\mathcal{E}_\theta \approx \mathcal{E}_N$, or separately evaluating both functionals, is an additional requirement. Accordingly, the experiments and code use distinct labels for learned-energy and physical-energy monotonicity.

REMARK 5.2 (REGULARITY HIERARCHY). Definition 5.2 assumes $\Psi_\theta \in C^1$, which suffices for the structural properties (Proposition 5.5). The global-flow existence result (Proposition 5.3) and the Lipschitz verification (Proposition 6.4) require the stronger regularity $\Psi_\theta \in C^2$ and $K_\theta \in C^1$, which is ensured by the neural parametrization of Section 6 when the activation ρ is C^2 (e.g., softplus or tanh).

PROPOSITION 5.3 (Sufficient conditions for global flow). Assume that $\Psi_\theta \in C^2(\mathbb{R}^N; \mathbb{R})$ satisfies the coercivity condition

$$(5.12) \quad \Psi_\theta(z) \geq c_1 \|z\|^2 - c_2, \quad z \in \mathbb{R}^N,$$

for some constants $c_1 > 0$ and $c_2 \geq 0$. Assume furthermore that $K_\theta \in C^1(\mathbb{R}^N; \mathbb{R}^{N \times N})$ is symmetric positive semidefinite for every $z \in \mathbb{R}^N$. Then the latent ODE (5.9) generates a global flow on $\mathbb{R}^N$.

Proof. Let $z_0 \in \mathbb{R}^N$ and let $Z_\theta(s; z_0)$ denote the maximal solution. Since $\Psi_\theta \in C^2$ and $K_\theta \in C^1$, the vector field $f_\theta(z) = -K_\theta(z) \nabla \Psi_\theta(z)$ is of class C^1 , hence locally Lipschitz on $\mathbb{R}^N$. The energy dissipation identity gives

$$\frac{d}{ds} \Psi_\theta(Z_\theta(s)) = -\nabla \Psi_\theta(Z_\theta(s))^\top K_\theta(Z_\theta(s)) \nabla \Psi_\theta(Z_\theta(s)) \leq 0,$$

so $\Psi_\theta(Z_\theta(s)) \leq \Psi_\theta(z_0)$ for all s in the maximal interval of existence. By coercivity (5.12), $\|Z_\theta(s)\|^2 \leq (\Psi_\theta(z_0) + c_2)/c_1$, so the solution remains bounded on any finite interval. By the extension theorem for ODEs, the maximal solution exists for all $s \geq 0$, i.e., the flow is global. $\square$

REMARK 5.3 (VERIFIABILITY OF THE GLOBAL-FLOW CONDITIONS). For the neural parametrization of Section 6, condition (5.12) holds if $\beta_V > 0$ in (6.9), since the quadratic term $\frac{\beta_V}{2} \|z\|^2$ dominates for large $\|z\|$. The regularity $K_\theta \in C^1$ and positive semidefiniteness hold automatically for the softplus-based mobility parametrization of (6.13). These conditions are therefore verifiable for concrete neural parametrizations and do not constitute additional abstract assumptions.

REMARK 5.4 (ARCHIVED NONCOERCIVE CONFIGURATION AND REVISED OPTION). The archived checkpoints reported in Section 7 were initialized with $\beta_V = 0$. They are therefore not covered by Proposition 5.3; absence of numerical blowup on finitely many trajectories is only a diagnostic and is not evidence of global existence. The revised implementation adds a non-trainable coefficient floor $\beta_{V,\min} \geq 0$ and uses the effective coefficient $\beta_V^{\text{eff}} = \beta_{V,\min} + |\tilde{\beta}_V|$. Selecting $\beta_{V,\min} > 0$ prevents optimization from removing the quadratic term and makes the coercivity configuration explicit. Results from such a configuration must be reported separately rather than attributed retroactively to archived checkpoints.

PROPOSITION 5.4 (Explicit global-flow criterion for the neural parametrization).

Consider the neural parametrization of Definitions 6.1–6.2. Suppose the following hold:

- (i) The quadratic coefficient satisfies $\beta_V > 0$ in (6.9).
- (ii) The MLP part $V_\theta^{\text{mlp}}(\xi) := (c^V)^\top h_{L_V-1}^V(\xi)$ satisfies a growth bound $|V_\theta^{\text{mlp}}(\xi)| \leq C_1(1 + |\xi|)$ for some $C_1 > 0$.
- (iii) The mobility coefficients are uniformly bounded: $k_{\theta,i}(z) \leq K_{\max}$ for all $z \in \mathbb{R}^N$ and $i = 1, \dots, N$, where $K_{\max} < \infty$ is a constant depending only on the network parameters.

Then the latent ODE (5.9) generates a global flow on $\mathbb{R}^N$.

Proof. We verify the hypotheses of Proposition 5.3. Condition (i) ensures $\Psi_\theta(z) \geq \frac{\beta_V}{2} \|z\|^2 - C_2$ for some $C_2 \geq 0$, which is the coercivity condition (5.12) with $c_1 = \beta_V/2$. Condition (ii) ensures the MLP component does not overwhelm the quadratic term at large $\|z\|$, preserving the coercivity of Ψ_θ . The mobility K_θ is symmetric positive semidefinite by construction (softplus output). The C^1 regularity of K_θ follows from the smooth activation $\rho = \text{softplus}$. The result then follows from Proposition 5.3. $\square$

PROPOSITION 5.5 (Structural properties of the latent-flow construction). Assume that (5.9) generates a global flow on $\mathbb{R}^N$. Then, for every $u \in (m, M)^N$ and every $\tau, \tau_1, \tau_2 \geq 0$, the map (5.10) is well defined, takes values in $(m, M)^N$, and satisfies

$$(5.13) \quad \Phi_\theta(u, 0) = u,$$

$$(5.14) \quad \Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) = \Phi_\theta(u, \tau_1 + \tau_2),$$

$$(5.15) \quad \Phi_\theta(u, \tau) \in \mathcal{K}_N,$$

and

$$(5.16) \quad \mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u).$$

Proof. Set $z_0 := g^{-1}(u)$. Equation (5.13) follows from the identity property of the flow. Equation (5.14) follows from the semigroup property of the autonomous

flow φ_θ^τ . Since $g(\mathbb{R}^N) \subset (m, M)^N \subset \mathcal{K}_N$, the map (5.10) is well defined for all $\tau \geq 0$, takes values in $(m, M)^N$, and in particular satisfies equation (5.15). Finally,

$$\begin{aligned} \frac{d}{ds} \Psi_\theta(Z_\theta(s; z_0)) &= \nabla \Psi_\theta(Z_\theta(s; z_0))^\top \partial_s Z_\theta(s; z_0) \\ &= -\nabla \Psi_\theta(Z_\theta(s; z_0))^\top K_\theta(Z_\theta(s; z_0)) \nabla \Psi_\theta(Z_\theta(s; z_0)) \\ &\leq 0. \end{aligned}$$

Hence (5.16) follows from (5.11). $\square$

PROPOSITION 5.6 (Generator expansion and matching condition). *Assume that $\Psi_\theta \in C^2(\mathbb{R}^N)$ and $K_\theta \in C^1(\mathbb{R}^N; \mathbb{R}^{N \times N})$. Then the latent vector field in (5.9) is of class C^1 , so it generates a local C^1 flow. For each fixed $u \in (m, M)^N$,*

$$(5.17) \quad \Phi_\theta(u, \tau) = u - \tau \text{D}g(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) + o(\tau), \quad \tau \downarrow 0.$$

Consequently, if there exists a semidiscrete generator $\mathcal{F}_N : (m, M)^N \rightarrow \mathbb{R}^N$ such that the matching condition

$$(5.18) \quad \text{D}g(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -\mathcal{F}_N(u), \quad u \in (m, M)^N,$$

is satisfied, then

$$\Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad \tau \downarrow 0,$$

which is the finite-dimensional counterpart of (3.3).

Proof. Fix $u \in (m, M)^N$, and set $z_0 := g^{-1}(u)$. Since the latent vector field

$$f_\theta(z) := -K_\theta(z) \nabla \Psi_\theta(z)$$

is C^1 , the local flow satisfies the standard first-order expansion

$$\varphi_\theta^\tau(z_0) = z_0 + \tau f_\theta(z_0) + o(\tau), \quad \tau \downarrow 0.$$

Because $g \in C^1(\mathbb{R}^N; (m, M)^N)$, Taylor expansion at z_0 gives

$$\begin{aligned} \Phi_\theta(u, \tau) &= g(\varphi_\theta^\tau(z_0)) \\ &= g(z_0) + \text{D}g(z_0)(\varphi_\theta^\tau(z_0) - z_0) + o(\|\varphi_\theta^\tau(z_0) - z_0\|) \\ &= u - \tau \text{D}g(z_0) K_\theta(z_0) \nabla \Psi_\theta(z_0) + o(\tau). \end{aligned}$$

This is (5.17). If (5.18) holds for some semidiscrete generator $\mathcal{F}_N$, then

$$\Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad \tau \downarrow 0,$$

which is the finite-dimensional counterpart of (3.3). $\square$

COROLLARY 5.7 (Verification of the finite-dimensional structural clauses). *Assume that the hypotheses of Proposition 5.5 hold. Then the map (5.10) satisfies (3.2) and (3.5) exactly on the interior state set $(m, M)^N$, it satisfies (3.4) there with defect function*

$$(5.19) \quad \eta_{\text{sg}}(\tau_1, \tau_2) \equiv 0.$$

and it satisfies the finite-dimensional dissipation law

$$(5.20) \quad \mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u), \quad u \in (m, M)^N, \quad \tau \geq 0.$$

If, in addition, Proposition 5.6 holds and

$$(5.21) \quad Dg(g^{-1}(u))K_\theta(g^{-1}(u))\nabla\Psi_\theta(g^{-1}(u)) = -\mathcal{F}_N(u), \quad u \in (m, M)^N,$$

then the finite-dimensional generator relation

$$(5.22) \quad \Phi_\theta(u, \tau) = u + \tau\mathcal{F}_N(u) + o(\tau), \quad u \in (m, M)^N,$$

holds as $\tau \downarrow 0$.

Consequently, the latent-flow construction provides an exact finite-dimensional verification of the identity, admissibility, exact semigroup, and induced-energy dissipation properties on the interior state set, while generator matching remains conditional on (5.21) and concerns a semidiscrete generator $\mathcal{F}_N$, not yet the original PDE generator $\mathcal{F}$.

Proof. Equation (3.2) follows from (5.13); equation (3.5) follows from (5.15); (3.4) follows from (5.14), which implies (5.19); and (5.20) follows from (5.16). The final assertion is immediate from Proposition 5.6 and (5.21). $\square$

REMARK 5.5 (CONTINUOUS-TIME VS. DISCRETE-TIME SEMIGROUP). The exact semigroup property (5.19) ($\eta_{\text{sg}} \equiv 0$) holds only for the exact continuous-time latent flow. A fourth-order Runge–Kutta implementation generally has a nonzero numerical composition defect. The archived value near 10^{-14} for $M_{\text{sub}} = 30$ is a floating-point observation for one protocol, not an exact guarantee and not by itself a verified integrator-error estimate. The revised evaluator reports absolute and relative numerical defects and supports substep refinement while restoring the model’s original substep count.

6. Neural parametrization of the latent semigroup. The previous section identifies the abstract ingredients needed for a structure-preserving semigroup learner. We now turn those ingredients into explicit neural parametrizations of the latent energy, interaction matrix, and mobility.

DEFINITION 6.1 (Reaction–diffusion latent energy ansatz). *Let*

$$(6.1) \quad V_\theta \in C^1(\mathbb{R}; \mathbb{R})$$

be a scalar potential, and let

$$(6.2) \quad a_{ij}^{(\theta)} = a_{ji}^{(\theta)} \geq 0, \quad i, j = 1, \dots, N.$$

Define

$$(6.3) \quad \Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2.$$

If, in addition,

$$(6.4) \quad K_\theta(z) = \text{diag}(k_{\theta,1}(z), \dots, k_{\theta,N}(z)), \quad k_{\theta,i}(z) \geq 0,$$

then the latent dynamics takes the componentwise form

$$(6.5) \quad \partial_s z_i = -k_{\theta,i}(z) \left(V'_\theta(z_i) + 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j) \right), \quad i = 1, \dots, N.$$

433 DEFINITION 6.2 (Neural parametrization of latent energy and mobility). *Fix an*
 434 *activation function $\rho : \mathbb{R} \rightarrow \mathbb{R}$ of class C^1 , and define*

$$435 \quad (6.6) \quad \text{softplus}(\xi) = \log(1 + e^\xi).$$

436 *For the scalar potential, fix integers $d_1^V, \dots, d_{L_V-1}^V \in \mathbb{N}$, together with parameters*

$$437 \quad (6.7) \quad W_0^V \in \mathbb{R}^{d_1^V \times 1}, \quad b_0^V \in \mathbb{R}^{d_1^V},$$

438

$$439 \quad W_\ell^V \in \mathbb{R}^{d_{\ell+1}^V \times d_\ell^V}, \quad b_\ell^V \in \mathbb{R}^{d_{\ell+1}^V}, \quad \ell = 1, \dots, L_V - 2, \quad c^V \in \mathbb{R}^{d_{L_V-1}^V}, \quad \beta_V \geq 0.$$

440 *Define*

$$441 \quad (6.8) \quad h_1^V(\xi) := \rho(W_0^V \xi + b_0^V), \quad h_{\ell+1}^V(\xi) := \rho(W_\ell^V h_\ell^V(\xi) + b_\ell^V), \quad \ell = 1, \dots, L_V - 2,$$

442 *and set*

$$443 \quad (6.9) \quad V_\theta(\xi) := (c^V)^\top h_{L_V-1}^V(\xi) + \frac{\beta_V}{2} \xi^2.$$

444 *For the interaction coefficients, fix an integer $r_A \geq 1$, choose symmetric binary*
 445 *weights $m_{ij} = m_{ji} \in \{0, 1\}$ and embeddings $e_i \in \mathbb{R}^{r_A}$ for $i = 1, \dots, N$, and define*

$$446 \quad (6.10) \quad a_{ij}^{(\theta)} := m_{ij} \text{softplus}(e_i^\top e_j), \quad i, j = 1, \dots, N.$$

447 *For the diagonal mobility, fix a stencil radius $r_K \in \mathbb{N}$ and integers $d_1^K, \dots, d_{L_K-1}^K \in$*
 448 *$\mathbb{N}$. Fix parameters*

$$449 \quad (6.11) \quad W_0^K \in \mathbb{R}^{d_1^K \times (2r_K+1)}, \quad b_0^K \in \mathbb{R}^{d_1^K},$$

450

$$451 \quad W_\ell^K \in \mathbb{R}^{d_{\ell+1}^K \times d_\ell^K}, \quad b_\ell^K \in \mathbb{R}^{d_{\ell+1}^K}, \quad \ell = 1, \dots, L_K - 2, \quad c^K \in \mathbb{R}^{d_{L_K-1}^K}, \quad \beta_K \in \mathbb{R}. \blacksquare$$

452 *For each $z \in \mathbb{R}^N$ and $i = 1, \dots, N$, extend boundary values by zero-padding (setting*
 453 *$z_j = 0$ for indices $j \notin \{1, \dots, N\}$) and define*

$$454 \quad (6.12) \quad h_{i,1}^K(z) := \rho(W_0^K(z_{i-r_K}, \dots, z_i, \dots, z_{i+r_K})^\top + b_0^K),$$

455

$$456 \quad h_{i,\ell+1}^K(z) := \rho(W_\ell^K h_{i,\ell}^K(z) + b_\ell^K), \quad \ell = 1, \dots, L_K - 2,$$

457 *and set*

$$458 \quad (6.13) \quad k_{\theta,i}(z) := \text{softplus}((c^K)^\top h_{i,L_K-1}^K(z) + \beta_K), \quad i = 1, \dots, N.$$

459 *Hence the diagonal mobility matrix is*

$$460 \quad (6.14) \quad K_\theta(z) := \text{diag}(k_{\theta,1}(z), \dots, k_{\theta,N}(z)).$$

461 The zero-padding convention is used only to make the finite-dimensional stencil
 462 map well defined. It can be replaced by another fixed extension rule, such as peri-
 463 odic, reflective, or boundary-condition-aware padding, without changing the abstract
 464 latent-flow argument above.

PROPOSITION 6.3 (Parameterized latent dynamics). *Under Definitions 6.1 and 6.2, $\Psi_\theta \in C^1(\mathbb{R}^N)$ and $K_\theta(z)$ is symmetric positive semidefinite for every $z \in \mathbb{R}^N$. These identities verify the regularity and positivity conditions in Definition 5.2. Moreover, the coefficients in (6.3) and (6.5) are realized by (6.10) and (6.13). Consequently,*

$$(6.15) \quad \Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2,$$

its gradient satisfies

$$(6.16) \quad (\nabla \Psi_\theta(z))_i = (V_\theta)'(z_i) + 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j), \quad i = 1, \dots, N,$$

and the latent ODE becomes

$$(6.17) \quad \partial_s z_i = -k_{\theta,i}(z) \left((V_\theta)'(z_i) + 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j) \right),$$

for $i = 1, \dots, N$.

Proof. By Definition 6.2, the scalar network V_θ is a finite composition of affine maps with the C^1 activation ρ , followed by the smooth quadratic term $\frac{\beta_V}{2} \xi^2$; hence $V_\theta \in C^1(\mathbb{R})$. Likewise, each coefficient $a_{ij}^{(\theta)} = m_{ij} \text{softplus}(e_i^\top e_j)$ is symmetric and nonnegative, and each mobility coefficient $k_{\theta,i}(z)$ is nonnegative because $\text{softplus} \geq 0$. Therefore $K_\theta(z)$ is diagonal with nonnegative diagonal entries, hence symmetric positive semidefinite for every $z \in \mathbb{R}^N$.

Since Ψ_θ in (6.3) is a finite sum of C^1 terms, it belongs to $C^1(\mathbb{R}^N)$. Differentiating (6.3) with respect to z_i and using the symmetry $a_{ij}^{(\theta)} = a_{ji}^{(\theta)}$ yields

$$\partial_{z_i} \left(\frac{1}{2} \sum_{p,q=1}^N a_{pq}^{(\theta)} (z_p - z_q)^2 \right) = 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j),$$

which gives (6.16). Substituting (6.16) and (6.13) into the latent ODE (5.9) gives (6.17). The opening assertions verify the regularity and positivity conditions appearing in Definition 5.2. $\square$

PROPOSITION 6.4 (One-step Lipschitz condition for the latent architecture). *Assume that the hypotheses of Proposition 5.3 hold and that the latent ODE generates a global flow. Fix $T > 0$ and a compact subset $\mathcal{U} \subset (m, M)^N$. If $\Psi_\theta \in C^2(\mathbb{R}^N)$ and $K_\theta \in C^1(\mathbb{R}^N; \mathbb{R}^{N \times N})$, then the latent vector field $f_\theta(z) = -K_\theta(z) \nabla \Psi_\theta(z)$ is locally Lipschitz on $\mathbb{R}^N$, and the flow φ_θ^τ is Lipschitz on the preimage $g^{-1}(\mathcal{U})$ with constant depending on T and $\mathcal{U}$. Consequently, there exists $L_T \geq 0$ such that*

$$(6.18) \quad \|\Phi_\theta(v, \tau) - \Phi_\theta(w, \tau)\| \leq (1 + L_T \tau) \|v - w\|, \quad v, w \in \mathcal{U}, \quad 0 \leq \tau \leq T.$$

In particular, the hypothesis (4.4) of Proposition 4.2 holds on compact subsets of the interior state set.

Proof. Since $\Psi_\theta \in C^2$ and $K_\theta \in C^1$, the vector field $f_\theta(z) = -K_\theta(z) \nabla \Psi_\theta(z)$ is C^1 , hence locally Lipschitz. On the compact set $g^{-1}(\mathcal{U})$, let L_f denote the Lipschitz constant of f_θ on a compact neighborhood. By the standard Lipschitz dependence of

ODE flows on initial data (obtained via Gronwall's inequality applied to the variational equation), the flow satisfies

$$\|\varphi_\theta^\tau(z_1) - \varphi_\theta^\tau(z_2)\| \leq e^{L_f \tau} \|z_1 - z_2\|, \quad z_1, z_2 \in g^{-1}(\mathcal{U}), \quad 0 \leq \tau \leq T.$$

The sigmoid map g is Lipschitz on $\mathbb{R}^N$ with constant $(M - m)/4$, and g^{-1} is Lipschitz on compact subsets of $(m, M)^N$ with constant $4/((M - m)\delta)$ where $\delta = \min_{u \in \mathcal{U}} \min_i \{u_i - m, M - u_i\} > 0$. Composing the three Lipschitz bounds gives the stated estimate with L_T depending on L_f , T , and the geometry of $\mathcal{U}$. $\square$

REMARK 6.1 (TIGHTNESS AT $\tau = 0$). At $\tau = 0$, the composition bound from the proof above gives $\|\Phi_\theta(v, 0) - \Phi_\theta(w, 0)\| \leq \|v - w\|/\delta$, which overestimates the true Lipschitz constant of $\Phi_\theta(\cdot, 0) = \text{Id}$ (which is exactly 1). The tighter form $\|\Phi_\theta(v, \tau) - \Phi_\theta(w, \tau)\| \leq (1 + L_T \tau) \|v - w\|$ stated in (6.18) follows from the mean value theorem applied to the Jacobian $D_v \Phi_\theta(v, \tau)$, which equals I at $\tau = 0$ and varies continuously with τ . This form preserves the linear-in- τ growth used in the Gronwall argument of Proposition 4.2.

7. Benchmark experiments. We illustrate how the abstract ingredients of Sections 5–6 apply to Fisher–KPP and Allen–Cahn reaction–diffusion equations. Viscous Burgers includes a transport term outside the pure latent gradient-flow model class and is included only as an empirical stress test. The numerical tables in this snapshot are archival: after correction of the reference-time alignment, every comparative value must be rerun before it is treated as revised evidence.

7.1. Semigroup-based training objectives. Suppose now that training data contain snapshots of the target evolution, namely triples $(u, \tau, \mathcal{S}_\tau u)$ with states sampled from the interior set on which the latent construction is defined. The training objective is the one-step prediction loss:

$$(7.1) \quad \mathcal{L}_{\text{step}} = \mathbb{E} \|\Phi_\theta(u, \tau) - \mathcal{S}_\tau u\|^2,$$

Training strategy in the experiments. The archived Fisher–KPP runner used $\mathcal{L}_{\text{step}} + 0.1\mathcal{L}_{\text{rollout}} + 0.01\mathcal{L}_{\text{learned energy}}$ for the latent model, so its accuracy cannot be described as obtained from the one-step loss alone. These auxiliary terms are not needed to define the exact continuous architecture, but they can affect optimization and empirical accuracy. The revised runner therefore records all loss weights and provides an architecture-only mode with both auxiliary weights set to zero. Learned-energy and physical-energy diagnostics are reported separately.

7.2. Fisher–KPP equation. Take the Fisher–KPP equation

$$(7.2) \quad \partial_t u = \nu \Delta u + ru(1 - u), \quad 0 \leq u \leq 1,$$

on a spatial domain $\Omega \subset \mathbb{R}^d$ with homogeneous Neumann or periodic boundary conditions. The admissible set is $\mathcal{K} = \{u : 0 \leq u \leq 1\}$ and the dynamics is the L^2 -gradient flow of the energy

$$(7.3) \quad \mathcal{E}(u) = \int_\Omega \left(\frac{\nu}{2} |\nabla u|^2 - r \left(\frac{u^2}{2} - \frac{u^3}{3} \right) \right) dx,$$

so that $\mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u)$ for all $t \geq 0$. The energy $\mathcal{E}$ is bounded from below on $\mathcal{K}$ since $r(u^2/2 - u^3/3) \leq r/6$ for $u \in [0, 1]$.

539 *Spatial discretization and admissible set..* Let $u = (u_1, \dots, u_N)^\top \in \mathbb{R}^N$ denote
 540 the spatially discretized state. The admissible set becomes the box

$$541 \quad (7.4) \quad \mathcal{K}_N = [0, 1]^N,$$

542 and the interior set on which the latent construction operates is $(0, 1)^N$.

543 *Latent map..* With $m = 0$, $M = 1$ the sigmoid parametrization of Definition 5.1
 544 simplifies to

$$545 \quad (7.5) \quad g(z)_i = \sigma(z_i), \quad g^{-1}(u)_i = \log \frac{u_i}{1 - u_i}, \quad i = 1, \dots, N.$$

546 *Latent energy and induced discrete energy..* The reaction–diffusion ansatz of Def-
 547 inition 6.1 takes the form

$$548 \quad (7.6) \quad \Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2,$$

549 where V_θ and $a_{ij}^{(\theta)}$ are parametrized as in Definition 6.2. The induced discrete energy
 550 on $(0, 1)^N$ is

$$551 \quad (7.7) \quad \mathcal{E}_\theta(u) = \Psi_\theta(g^{-1}(u)).$$

552 *Architecture guarantees..* By Proposition 5.5, assuming the latent ODE generates
 553 a global flow (which holds under the conditions of Proposition 5.3), the learned fam-
 554 ily $\Phi_\theta(\cdot, \tau)$ satisfies admissibility ($\Phi_\theta(u, \tau) \in [0, 1]^N$) and induced-energy dissipation
 555 $\mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u)$ exactly on $(0, 1)^N$.

556 *Matching condition..* Let Δ_h denote the discrete Laplacian corresponding to the
 557 spatial discretization and $\odot$ denote componentwise multiplication. The generator
 558 matching condition (5.18) then reads

$$559 \quad (7.8) \quad \text{D}g(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -(\nu \Delta_h u + r u \odot (1 - u)), \quad u \in (0, 1)^N.$$

560 This identity is a target to be realized by the neural parametrization of Ψ_θ and K_θ ;
 561 it is not automatically satisfied by the architecture.

562 **7.3. Allen–Cahn equation.** As a second benchmark, consider the Allen–Cahn
 563 equation

$$564 \quad (7.9) \quad \partial_t u = \varepsilon^2 \Delta u + u(1 - u^2), \quad -1 \leq u \leq 1,$$

565 on $\Omega \subset \mathbb{R}^d$ with homogeneous Neumann or periodic boundary conditions. The ad-
 566 missible set is $\mathcal{K} = \{u : -1 \leq u \leq 1\}$, and the associated energy is

$$567 \quad (7.10) \quad \mathcal{E}(u) = \int_\Omega \left(\frac{\varepsilon^2}{2} |\nabla u|^2 + \frac{(1 - u^2)^2}{4} \right) dx,$$

568 which satisfies $\mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u)$ for all $t \geq 0$. The admissible set becomes $\mathcal{K}_N = [-1, 1]^N$
 569 with interior $(-1, 1)^N$. The latent map, energy ansatz, and matching condition follow
 570 the same pattern as for Fisher–KPP with $m = -1$, $M = 1$, and the generator matching
 571 target

$$572 \quad (7.11) \quad \text{D}g(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -(\varepsilon^2 \Delta_h u + u \odot (1 - u \odot u)).$$

7.4. Experimental setup.

Architecture details. Table 1 summarizes the network architectures used in all experiments. The latent learner uses the parametrization of Definitions 6.1–6.2 with a local interaction mask (each site connected to its $2r_K$ nearest neighbors, periodic padding for K_θ , periodic interaction graph for $a_{ij}^{(\theta)}$). The activation function ρ is softplus throughout, which is C^∞ and compatible with the regularity requirements of Propositions 5.3–6.4. The ResNet baseline uses a standard residual network with skip connections, trained with a pointwise MSE loss.

TABLE 1
Architecture details for the latent semigroup learner, FNO, and ResNet baseline.

Component	Latent learner	FNO	ResNet baseline
Potential V_θ	$L_V=2$, widths (64, 64), $\beta_V=0^\dagger$	—	—
Interaction rank r_A	8 (nearest-neighbor, radius 2)	—	—
Mobility $k_{\theta,i}$	$L_K=2$, widths (64, 64), $r_K=3$	—	—
Fourier layers	—	4 layers, 16 modes, width 16	—
Residual blocks	—	—	4 blocks, 32 ch.
Total parameters	~9.6 K	~17.5 K	~41.6 K
Optimizer	AdamW, wd= 10^{-5}	AdamW, wd= 10^{-5}	AdamW, wd= 10^{-5}
Grad clip	1.0	1.0	1.0
Learning rate	10^{-3} (cosine decay)	10^{-3} (cosine decay)	10^{-3} (cosine decay)
Batch size	64	64	64
Training epochs	100	100	100

[†]Archived experiments use $\beta_V = 0$ and are not covered by Proposition 5.3; finite-trajectory stability is only an empirical diagnostic (Remark 5.4).

Training data. For each benchmark, training data consist of triples $(u_0, \tau, \mathcal{S}_\tau u_0)$ with 1000 training samples and 50 hold-out test samples. The initial conditions u_0 are drawn from PDE-specific distributions within the interior of the admissible set: for Fisher–KPP, $u_0 \sim \text{Uniform}[0.3, 0.7]^N$; for Allen–Cahn, $u_0 = \tanh((x-0.5)/(\sqrt{2}\varepsilon)) + \xi$ with $\xi \sim 0.05\mathcal{N}(0, I_N)$; for Burgers, $u_0 = \sin(2\pi x/L) \cdot (0.5 + 0.5\eta) + \xi$ with $\eta \sim \text{Uniform}[0, 1]$ and $\xi \sim 0.1\mathcal{N}(0, I_N)$. The time step is $\tau = \Delta t = 0.1$ for Fisher–KPP and Allen–Cahn, and $\tau = \Delta t = 0.05$ for the stiffer Burgers benchmark. The reference solution $\mathcal{S}_\tau u_0$ is computed by a high-resolution numerical integrator with a finer internal time step Δt_{ref} : central finite differences with RK4 ($\Delta t_{\text{ref}} = 0.005$) for Fisher–KPP, upwind finite differences with RK4 ($\Delta t_{\text{ref}} = 0.001$) for Burgers, and a semi-implicit IMEX scheme with spectral treatment of the diffusion term ($\Delta t_{\text{ref}} = 0.0005$) for Allen–Cahn.

Computational cost. All experiments were conducted on a single NVIDIA GeForce RTX 5080 GPU. The latent learner requires $M_{\text{sub}} = 30$ RK4 substeps per time step, corresponding to 120 function evaluations per step (vs. 1 for FNO and ResNet). Despite this overhead, the latent learner’s total training time is comparable to the baselines due to its smaller parameter count (9.6 K vs. 17.5 K for FNO and 41.6 K for ResNet). The per-epoch training times are approximately 45 s for the latent learner, 30 s for FNO, and 35 s for ResNet on the Fisher–KPP benchmark ($N = 64$, batch size 64).

Reproducibility. The archived runs did not fix every source of randomness. The revised runners seed Python, NumPy, PyTorch, and CUDA from one recorded seed and optionally request deterministic PyTorch algorithms. Linear latent weights retain the initialization $\mathcal{N}(0, 0.001)$ with zero biases. Validation-trajectory variation is not

a substitute for variation across independently trained seeds; revised comparative claims should therefore report multiple training seeds or explicitly identify a single-seed result.

7.5. Results.

Main comparison (Fisher–KPP). Table 2 reports the evaluation metrics on 50 hold-out initial conditions, each rolled out over 20 time steps ($T_{\max} = 2.0$).

TABLE 2

Archived Fisher–KPP results (50 test samples, mean $\pm$ std). The rollout values were produced before strict reference-time alignment was enforced and are retained for provenance, not as revised comparative evidence.

Metric	Latent semigroup	FNO	ResNet baseline
Parameters	9.6 K	17.5 K	41.6 K
Rollout MSE (final)	$(1.49 \pm 0.15) \times 10^{-4}$	$(2.10 \pm 1.07) \times 10^{-3}$	$(2.86 \pm 0.22) \times 10^{-2}$
Boundary violation	0.00 ¹	$(4.56 \pm 12.2) \times 10^{-4}$	$(7.23 \pm 11.52) \times 10^{-4}$
Numerical SG defect (archived MSE)	$\sim 10^{-14}$	$(2.19 \pm 0.83) \times 10^{-5}$	$(2.05 \pm 0.15) \times 10^{-4}$
Learned-energy monotonicity	100.0% (archived diagnostic)	Not evaluated	Not evaluated

The archived table reported a roughly 190-fold MSE ratio, but the former evaluator advanced the model by $\tau = 0.1$ while indexing the dense Fisher–KPP reference trajectory as though adjacent snapshots were separated by the same interval; they were separated by 0.005. The ratio must therefore be recomputed. The zero bound-violation observation remains consistent with the bounded decoder on finite numerical trajectories, but the archived $\beta_V = 0$ configuration is outside the stated coercive global-flow theorem. Likewise, the 100% value concerns $\mathcal{E}_\theta$, not the physical energy (7.3), and the reported composition residual is numerical.

Convergence under spatial refinement. Table 3 reports the final-step trajectory MSE for the latent learner on Fisher–KPP as the spatial resolution N varies. The error decreases monotonically with N , consistent with the increasing expressiveness of the latent architecture. The successive refinement ratios are approximately $7.2 \times$ ($N=16 \rightarrow 32$), $8.1 \times$ ($N=32 \rightarrow 64$), and $3.9 \times$ ($N=64 \rightarrow 128$). The first two doublings are consistent with an effective convergence rate of approximately third order in $1/N$ (since $\log_2(7.5) \approx 2.9$), while the reduced ratio at $N=128$ suggests that the optimizer or expressiveness ceiling begins to limit further improvement. A rigorous convergence-rate analysis is deferred to Problem 10.1.

TABLE 3

Convergence under spatial refinement for the latent learner on Fisher–KPP ($T_{\max} = 2.0$, $\Delta t = 0.1$; results are from the main experiment code with N varied in the configuration).

N	Rollout MSE (final step)
16	$(8.72 \pm 1.03) \times 10^{-3}$
32	$(1.21 \pm 0.18) \times 10^{-3}$
64	$(1.49 \pm 0.15) \times 10^{-4}$
128	$(3.84 \pm 0.42) \times 10^{-5}$

REMARK 7.1 (SCOPE OF THE CONVERGENCE STUDY). The convergence study (Table 3) is presented only for Fisher–KPP. For Allen–Cahn and Burgers, the same latent architecture is used with identical hyperparameters; we defer PDE-specific convergence analysis to future work. A temporal convergence study (varying Δt

at fixed N) is also deferred, as the current experiments fix Δt based on the PDE timescale.

Long-time time-stepping.. Table 4 reports the trajectory MSE at $T = 10.0$ (100 time steps) and $T = 20.0$ (200 time steps) for Fisher-KPP with $N = 64$, comparing the latent learner, FNO, and ResNet baseline. The ResNet baseline exhibits numerical blowup beyond $T = 10$, with MSE reaching $\sim 10^{16}$ at $T = 20$. The FNO maintains finite error but with growing boundary violations (1.8% at $T = 20$). The latent learner is the only model that maintains both bounded error and exact boundary compliance throughout.

TABLE 4

Long-time rollout comparison on Fisher-KPP ($N = 64$, $\Delta t = 0.1$). The ResNet baseline exhibits numerical blowup beyond $T = 10$. The $T=2$ row reproduces the short-time comparison from Table 2 (50 test samples) for reference; the $T=10$ and $T=20$ rows are from a separate long-time rollout experiment (20 test samples).

$T_{\max}$ (steps)	Latent semigroup	FNO	ResNet
2.0 (20)	$(1.49 \pm 0.15) \times 10^{-4}$	$(2.10 \pm 1.07) \times 10^{-3}$	$(2.86 \pm 0.22) \times 10^{-2}$
10.0 (100)	$(4.09 \pm 0.14) \times 10^{-1}$	$(4.52 \pm 0.34) \times 10^{-1}$	1.56 ± 0.96
20.0 (200)	$(4.10 \pm 0.15) \times 10^{-1}$	$(4.66 \pm 0.33) \times 10^{-1}$	$(8.60 \pm 22.0) \times 10^{16}$

Boundary violation growth.. Table 5 reports the mean boundary violation at selected time horizons. The latent learner maintains zero violation at all times by construction. The FNO and ResNet baseline both exhibit growing violations; the ResNet baseline eventually leaves the admissible set entirely.

TABLE 5

Boundary violation growth on Fisher-KPP ($N = 64$, $\Delta t = 0.1$; 20 test samples from the long-time rollout experiment).

T	Latent semigroup	FNO	ResNet
0.5	0.00	$(8.8 \pm 2.1) \times 10^{-6}$	0.00
2.0	0.00	$(1.8 \pm 0.5) \times 10^{-3}$	$(1.3 \pm 0.4) \times 10^{-3}$
10.0	0.00	$(1.5 \pm 0.3) \times 10^{-2}$	$(6.0 \pm 1.1) \times 10^{-1}$
20.0	0.00	$(1.8 \pm 0.4) \times 10^{-2}$	$(8.7 \pm 2.3) \times 10^7$

Allen-Cahn benchmark.. The Allen-Cahn rerun protocol uses $\varepsilon = 0.1$, $N = 64$, $\Delta t = 0.1$, and $T_{\max} = 2.0$. Complete revised metrics are not available in this snapshot, so Table 6 records their status rather than displaying fabricated placeholders.

TABLE 6

Status of the revised Allen-Cahn evaluation ($\varepsilon = 0.1$, $N = 64$, $\Delta t = 0.1$, $T_{\max} = 2.0$).

Metric	Latent semigroup	FNO	ResNet baseline
Rollout MSE (final step)	Not reported	Not reported	Not reported
Boundary violation	Not reported	Not reported	Not reported
Numerical SG defect	Not reported	Not reported	Not reported
Learned / physical energy	Not reported	Not evaluated	Not evaluated

Discussion.. The archived tables motivate the revised study but do not yet establish comparative accuracy under the corrected evaluator. The bounded decoder gives a clear finite-trajectory constraint mechanism, while claims about PDE accu-

racy, physical-energy decay, numerical composition error, or spatial rates require the aligned multi-seed reruns described in the accompanying revision plan.

7.6. Viscous Burgers equation. As an out-of-class transport stress test, consider the viscous Burgers equation

$$(7.12) \quad \partial_t u + u \partial_x u = \nu \partial_{xx} u, \quad -1 \leq u \leq 1,$$

on $[0, 2\pi]$ with periodic boundary conditions and $\nu = 0.01$. The admissible set is $\mathcal{K}_N = [-1, 1]^N$, and the energy $\mathcal{E}(u) = \frac{1}{2} \int u^2 dx$ satisfies $\mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u)$ for all $t \geq 0$, since the convective term $\int u u_x u dx = \frac{1}{3} \int \partial_x(u^3) dx = 0$ under periodic boundary conditions. This benchmark is moderately stiffer than Fisher–KPP: the smaller viscosity $\nu = 0.01$ permits near-shock formation, and the nonlinear convection term $u \partial_x u$ introduces genuine transport dynamics absent from the reaction–diffusion benchmarks. The Burgers benchmark uses $\Delta t = 0.05$ (versus $\Delta t = 0.1$ for Fisher–KPP) to accommodate the smaller viscosity coefficient.

The physical L^2 energy dissipates under periodic boundary conditions, but the convective term is not a pure gradient contribution of the implemented form. Consequently, neither the transfer theorem nor the expressivity result is claimed for this benchmark. Existing FNO and ResNet values are retained as archived baselines; the missing latent entries are reported as unavailable.

TABLE 7
Archived baselines and current status for the out-of-class viscous Burgers stress test ($\nu = 0.01$, $N = 64$, $\Delta t = 0.05$, $T_{\max} = 1.0$).

Metric	Latent semigroup	FNO	ResNet
Rollout MSE (final)	Not reported	$(5.31 \pm 3.96) \times 10^{-2}$	$(8.19 \pm 2.51) \times 10^{-2}$
Boundary violation	Not reported	$(1.54 \pm 0.54) \times 10^{-1}$	$(6.56 \pm 2.00) \times 10^{-2}$
Numerical SG defect (archived MSE)	Not reported	$(2.66 \pm 1.98) \times 10^{-4}$	$(1.30 \pm 0.31) \times 10^{-3}$
Learned / physical energy	Not reported	Not evaluated	Not evaluated

Long-time Burgers rollout and an architecture with an explicit skew transport component are deferred to future work.

7.7. Ablation study. To isolate the contribution of each architectural component, we evaluate three variants of the latent semigroup learner on the Fisher–KPP benchmark (Table 8). The *full* model uses the complete architecture of Definitions 6.1–6.2. The *no-interaction* variant sets the interaction coefficients $a_{ij}^{(\theta)} = 0$ in (6.3), removing spatial coupling. The *no-mobility* variant replaces the learned diagonal mobility $K_\theta(z)$ with the identity matrix, reducing the latent dynamics to a plain gradient flow.

In the archived ablation, removing spatial interaction changed the reported Fisher–KPP error little, whereas replacing the learned mobility produced a roughly 36-fold error ratio. These observations are hypotheses for the aligned rerun, not causal conclusions: the variants must be trained across matched seeds before attributing the difference to a component. All finite numerical trajectories remained inside the decoder range.

8. Continuous-to-discrete structural transfer. This section keeps spatial discretization, exact finite-dimensional flows, and numerical time integration as three distinct objects. Let $P_N : X \rightarrow X_N$ be a projection or sampling map and let $I_N :$

TABLE 8

Ablation study on Fisher-KPP ($N = 64$, $\Delta t = 0.1$, $T_{\max} = 2.0$). All variants maintain exact boundary compliance by construction (under the global-flow assumption of Proposition 5.3; see the footnote on Table 2 for the guarantee). The full-architecture MSE (1.89×10^{-4}) differs from Table 2 (1.49×10^{-4}) due to different random seeds governing the training-validation split and parameter initialization.

Variant	Rollout MSE	Boundary violation
Full architecture	$(1.89 \pm 0.18) \times 10^{-4}$	0.00
No interaction ($a_{ij} = 0$)	$(1.89 \pm 0.18) \times 10^{-4}$	0.00
No mobility ($K = I$)	$(6.81 \pm 4.77) \times 10^{-3}$	0.00

689 $X_N \rightarrow X$ be a reconstruction map. We write $\tilde{S}_t^N$ for the exact flow generated by

690 (8.1) $\dot{v} = \mathcal{F}_N(v), \quad v(0) = v_0 \in X_N,$

691 and Q_h^N for a numerical one-step approximation to that flow. Neither symbol is used
692 for both objects.

693 ASSUMPTION 8.1 (EXACT SEMIDISCRETE APPROXIMATION). *Fix $T > 0$ and a*
694 *compact set $\mathcal{U} \subset \mathcal{K}$. Assume that $P_N\mathcal{U}$ is contained in a compact set $\mathcal{U}_N \subset (m, M)^N$*
695 *and that:*

- 696 (i) $\mathcal{F}_N$ is locally Lipschitz and is L_N -Lipschitz on a neighborhood containing both
697 $\tilde{S}_t^N(P_N\mathcal{U})$ and $\Phi_\theta(P_N\mathcal{U}, t)$ for $0 \leq t \leq T$;
698 (ii) the reconstruction is stable, $\|I_N v\|_X \leq C_I \|v\|_N$;
699 (iii) the exact semidiscrete flow converges after reconstruction:

700 (8.2)
$$\sup_{u_0 \in \mathcal{U}} \sup_{0 \leq t \leq T} \|I_N \tilde{S}_t^N(P_N u_0) - \mathcal{S}_t u_0\|_X \leq \varepsilon_{\text{space}}(N),$$

701 where $\varepsilon_{\text{space}}(N) \rightarrow 0$ as $N \rightarrow \infty$.

702 The PDE-specific projection, reconstruction, norm, invariant neighborhood, and proof
703 or citation for (8.2) must be supplied when the theorem is applied.

704 Define the observable-space generator of the exact latent flow by

705 (8.3)
$$F_\theta(v) := -\text{D}g(g^{-1}(v))K_\theta(g^{-1}(v))\nabla\Psi_\theta(g^{-1}(v)).$$

706 THEOREM 8.1 (Conditional exact-flow transfer). *Under Assumption 8.1, suppose*
707 *the exact latent flow exists on $[0, T]$ for the stated initial data and*

708 (8.4)
$$\sup_{v \in \mathcal{U}_N} \|F_\theta(v) - \mathcal{F}_N(v)\|_N \leq \varepsilon_{\text{match}}.$$

709 Let

710
$$\chi_L(t) := \begin{cases} (e^{Lt} - 1)/L, & L > 0, \\ t, & L = 0. \end{cases}$$

711 Then, for $u_0 \in \mathcal{U}$ and $0 \leq t \leq T$,

712 (8.5)
$$\|\Phi_\theta(P_N u_0, t) - \tilde{S}_t^N(P_N u_0)\|_N \leq \varepsilon_{\text{match}} \chi_{L_N}(t),$$

713 and consequently

714 (8.6)
$$\|I_N \Phi_\theta(P_N u_0, t) - \mathcal{S}_t u_0\|_X \leq C_I \varepsilon_{\text{match}} \chi_{L_N}(t) + \varepsilon_{\text{space}}(N).$$

715 The admissibility and learned-energy statements of Proposition 5.5 remain exact finite-
 716 dimensional properties of Φ_θ ; equation (8.6) does not identify $\mathcal{E}_\theta$ with the PDE energy
 717 $\mathcal{E}$.

718 *Proof.* Set $y(t) = \Phi_\theta(P_N u_0, t)$ and $x(t) = \tilde{S}_t^N(P_N u_0)$. The two curves have the
 719 same initial value. Adding and subtracting $\mathcal{F}_N(y(t))$, using (8.4) and the L_N -Lipschitz
 720 property of $\mathcal{F}_N$, gives

$$721 \quad \|y(t) - x(t)\|_N \leq \int_0^t (\varepsilon_{\text{match}} + L_N \|y(s) - x(s)\|_N) ds.$$

722 Continuous Gronwall yields (8.5). Reconstruction stability, the triangle inequality,
 723 and (8.2) then give (8.6). $\square$

724 PROPOSITION 8.2 (Adding numerical time-integration errors). *Let $t = nh \leq$
 725 T . Suppose $(Q_h^N)^n$ approximates $\tilde{S}_t^N$ with uniform error $\eta_N(h, T)$ and a numerical
 726 implementation $\hat{\Phi}_{\theta, h}^n$ approximates the exact latent flow with uniform error $\eta_\theta(h, T)$
 727 on the relevant initial states. Then*

$$728 \quad (8.7) \quad \left\| I_N \hat{\Phi}_{\theta, h}^n(P_N u_0) - \mathcal{S}_t u_0 \right\|_X \leq C_I \eta_\theta(h, T) + C_I \varepsilon_{\text{match}} \chi_{L_N}(t) + \varepsilon_{\text{space}}(N),$$

$$729 \quad (8.8) \quad \left\| I_N (Q_h^N)^n(P_N u_0) - \mathcal{S}_t u_0 \right\|_X \leq C_I \eta_N(h, T) + \varepsilon_{\text{space}}(N).$$

731 *Proof.* Insert the corresponding exact finite-dimensional flow in each expression
 732 and apply the triangle inequality, Theorem 8.1, and reconstruction stability. $\square$

733 REMARK 8.1 (NUMERICAL SEMIGROUP DEFECT). For a finite-step implementation,
 734 the measured composition residual

$$735 \quad \left\| \hat{\Phi}_{\theta, h}(\hat{\Phi}_{\theta, h}(v, s), t) - \hat{\Phi}_{\theta, h}(v, s + t) \right\|$$

736 is a *numerical semigroup defect*. It is not the error η_θ itself and is not an exact-flow
 737 theorem. The exact autonomous latent flow has zero composition defect, whereas
 738 RK4 generally does not. Substep-refinement experiments are therefore needed before
 739 attributing a measured residual to integrator accuracy.

740 **9. Expressivity within the implemented architecture.** The implemented
 741 family uses a shared scalar potential, a prescribed interaction graph with Gram-
 742 constrained coefficients, and local diagonal mobility. These restrictions preclude an
 743 unqualified density claim for arbitrary smooth gradient fields. The following result
 744 records what follows rigorously once component approximation has been established
 745 for an architecture-compatible target.

746 PROPOSITION 9.1 (Componentwise generator error). *Let $Z \subset \mathbb{R}^N$ be compact*
 747 *and suppose an architecture-compatible target has the form*

$$748 \quad F^*(g(z)) = -Dg(z)K^*(z)q^*(z),$$

749 *where $q^* = \nabla \Psi^*$ has the shared-potential and prescribed-interaction form of (6.16),*
 750 *and K^* has the permitted local diagonal form. Assume on Z that*

$$751 \quad \|Dg(z)\| \leq M_g, \quad \|K^*(z)\| \leq M_K, \quad \|q^*(z)\| \leq M_\Psi,$$

752 *and that one parameter choice satisfies*

$$753 \quad \sup_Z \|K_\theta - K^*\| \leq \varepsilon_K, \quad \sup_Z \|\nabla \Psi_\theta - q^*\| \leq \varepsilon_\Psi.$$

754 Then

$$755 \quad (9.1) \quad \sup_{z \in Z} \|F_\theta(g(z)) - F^*(g(z))\| \leq M_g (M_K \varepsilon_\Psi + M_\Psi \varepsilon_K + \varepsilon_K \varepsilon_\Psi).$$

756 *Proof.* Add and subtract $Dg(z)K^*(z)\nabla\Psi_\theta(z)$ and use submultiplicativity. The
757 bound $\|\nabla\Psi_\theta\| \leq M_\Psi + \varepsilon_\Psi$ gives (9.1). $\square$

758 REMARK 9.1 (SCOPE OF THE EXPRESSIVITY RESULT). Proposition 9.1 is an error-
759 propagation result, not a universal approximation theorem. Simultaneous density
760 under the shared scalar potential, finite-rank Gram interactions, fixed graph, and
761 local diagonal mobility remains open. In particular, the transport term in viscous
762 Burgers is not a pure gradient flow in this model class; Burgers is therefore an out-of-
763 class empirical stress test. A natural extension is a metriplectic or port-Hamiltonian
764 decomposition $\dot{z} = (J_\theta - K_\theta)\nabla\Psi_\theta$ with $J_\theta^\top = -J_\theta$.

765 COROLLARY 9.2 (Conditional architecture-compatible approximation). *If, along*
766 *a sequence of model classes, the two component errors in Proposition 9.1 converge to*
767 *zero and Assumption 8.1 holds, then the exact latent flow converges to the PDE*
768 *semigroup according to (8.6). Any numerical implementation additionally carries the*
769 *explicit term $\eta_\theta(h, T)$ from Proposition 8.2.*

770 **10. Research agenda and open problems.** The formulation above separates
771 four analytical tasks that are often conflated in one-step operator learning. The first
772 is an approximation question for nonlinear semigroups. The second is an architec-
773 ture question: which structural properties can be enforced exactly by the model class
774 itself? The third is a long-time composition question, concerned with how local ap-
775 proximation and semigroup defects accumulate under repeated composition. The
776 fourth is a transfer question, asking how continuous PDE-level structure is inherited
777 by finite-dimensional semidiscrete models.

778 PROBLEM 10.1 (SEMIGROUP EXPRESSIVITY). *Establish simultaneous component-*
779 *density results for the exact shared-potential, Gram-constrained interaction, fixed-*
780 *graph, and local-mobility classes. Proposition 9.1 propagates component errors but*
781 *does not prove that these errors can be driven to zero for arbitrary gradient-type*
782 *generators; extending the architecture to non-gradient generators is a further problem.*

783 PROBLEM 10.2 (EXACT STRUCTURAL REALIZATION). *Characterize neural operator*
784 *architectures for which admissibility and dissipation are exact model-class properties,*
785 *rather than penalties enforced only through training data. For such architectures,*
786 *analogues of (3.5) and (3.6) should hold for all admissible parameters on the stated*
787 *state set.*

788 PROBLEM 10.3 (ROLLOUT STABILITY UNDER COMPOSITION DEFECTS). *Develop*
789 *long-time error estimates for learned evolution families whose semigroup law is only*
790 *approximate. Given a stability mechanism for repeated time-stepping, quantify how*
791 *trajectory-level approximation error and semigroup defect accumulate under composi-*
792 *tion, and determine sharp analogues of (4.7) beyond the Lipschitz setting.*

793 PROBLEM 10.4 (CONTINUOUS-TO-DISCRETE STRUCTURAL TRANSFER). *Relate the*
794 *PDE-level maximum principle and Lyapunov dissipation law of $(S_t)_{t \geq 0}$ to finite-*
795 *dimensional admissible sets, discrete energies, and generator-matching conditions.*
796 *Such a result would provide the missing bridge between the continuous structural spec-*
797 *ification in Definition 3.1 and the semidiscrete latent-flow guarantees of Sections 5–6.*
798 *Theorem 8.1 provides a first result in this direction under the assumption of a consis-*
799 *tent semidiscrete discretization.*

11. Conclusion. This paper formulates structure-preserving semigroup learning for nonlinear evolution PDEs and establishes a layered set of conditional results.

First, we derive a discrete stability analysis (Proposition 4.2) that reveals how one-step approximation error and semigroup defect combine under repeated composition. This analysis (Section 4) shows that the semigroup defect enters the long-time error bound with a factor of n (the number of composition steps), making semigroup consistency a first-order concern for long-time rollout, not merely a formal property.

Second, we construct a semidiscrete latent dissipative architecture (Sections 5–6) for which invariance of the interior state set $(m, M)^N$, exact semigroup composition, and dissipation of an induced discrete energy are *exact* architectural guarantees—holding for every parameter θ and requiring no training penalty—provided the latent ODE generates a global flow (Proposition 5.3). These guarantees are verified by construction at the finite-dimensional level and are independent of training quality.

Third, the revised Transfer Theorem (Theorem 8.1) compares the exact latent and exact semidiscrete flows by continuous Gronwall, with explicit projection, reconstruction, norm, and spatial-convergence hypotheses. Proposition 8.2 adds numerical time-integration errors without calling them spatial errors or exact semigroup defects. Proposition 9.1 then gives a componentwise generator bound for architecture-compatible targets; it is deliberately not a universal approximation theorem.

Gap between architecture-level and PDE-level guarantees. The finite-dimensional guarantees concern the exact latent flow and the induced learned energy. They do not imply exact properties of RK4, decay of the physical PDE energy, or PDE accuracy without the hypotheses of Theorem 8.1. Component density remains open even within the restricted gradient class (Problem 10.1), while transport-dominated systems such as Burgers require an extended generator class. The archived $\beta_V = 0$ runs are also outside the coercive global-flow theorem; the revised code exposes a fixed positive floor for new theorem-covered runs.

Directions for future work. Several concrete extensions merit investigation:

- (1) **Two-dimensional spatial domains.** The current architecture is formulated for one-dimensional spatial domains with local nearest-neighbor coupling. Extending to two-dimensional domains requires a spatial interaction structure (e.g., graph neural network coupling or convolutional mobility) that preserves the dissipative gradient-flow structure while scaling to larger state dimensions.
- (2) **Tighter error bounds.** The Transfer Theorem bound scales linearly with T through the factor $e^{L_T T}$. Sharper estimates exploiting the dissipative structure of the latent flow—for example, contractivity of the linearized flow or exponential decay of perturbations—could yield bounds that do not grow (or grow more slowly) with the time horizon.
- (3) **Non-gradient generators.** Adding a skew component, for example $\dot{z} = (J_\theta - K_\theta)\nabla\Psi_\theta$ with $J_\theta^\top = -J_\theta$, would broaden applicability to PDEs with advection, transport, or non-symmetric dynamics.
- (4) **Adaptive time stepping.** The current formulation uses a fixed step size Δt . An adaptive scheme that varies Δt along the trajectory—exploiting the continuous-time latent flow—could improve efficiency for stiff or multiscale problems.
- (5) **Rigorous convergence rates.** The archived resolution study suggests decreasing error, but an aligned multi-seed rerun and a proof relating spatial resolution, network width, optimization, and PDE regularity are still required.

The framework therefore separates what can be guaranteed exactly at the archi-

texture level from what must be addressed through further analysis, and the research agenda laid out in Section 10 provides a concrete roadmap for closing the gap between finite-dimensional structural guarantees and PDE-level semigroup approximation.

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